

Health Manager - Console-Based Health Tracking Application

Overview

Health Manager is a comprehensive Python-based console application designed to help users track their health and fitness journey. This application provides a simple yet powerful solution for monitoring daily meals, exercise routines, and overall health progress through an intuitive text-based interface. With secure local data storage and no external dependencies, it's perfect for individuals who want to maintain their health records privately and efficiently.

Features

User Management

- **Secure Registration & Login** - Multi-user system with individual data isolation
- **Persistent Sessions** - Maintain user data across application restarts
- **Privacy Focused** - Each user's data stored separately in JSON files

Health Tracking

- **Meal Logging** - Record food consumption with calorie information
- **Exercise Monitoring** - Track workouts with duration and calories burned
- **Flexible Timestamps** - Automatic or custom date/time recording
- **Data Validation** - Input validation for calories and numerical data

Analytics & Reports

- **Comprehensive Summaries** - View meal and exercise history with totals
- **Calorie Analysis** - Net calorie calculation (intake vs burned)
- **Progress Tracking** - Records sorted by date with newest first
- **Health Insights** - Calorie surplus/deficit status indicators

Data Management

- **JSON Storage** - Structured data storage using human-readable JSON files
- **Data Persistence** - All records maintained between sessions
- **Backup Friendly** - Easy to backup and restore user data files

Technologies Used

- **Python 3.6+** - Core programming language
- **JSON** - Data serialization and storage
- **datetime** - Timestamp and date handling
- **os module** - File system operations
- **Standard Library Only** - No external dependencies required

Installation & Setup

Prerequisites

- Python 3.6 or higher installed on your system
- Basic terminal/command prompt knowledge

Step-by-Step Installation

1. Download the Project

Clone the repository or download the health_manager.py file

```
git clone <repository-url>
```

```
cd health-manager
```

2. Verify Python Installation

```
python --version
```

Should show Python 3.6 or higher

3. Run the Application

```
python health_manager.py
```

4. First-Time Setup

- The application will automatically create a users directory
- No additional configuration required

Project Structure

text

health-manager/

|

|— health_manager.py # Main application file

|— README.md # Project documentation

└─ users/ # User data directory (auto-created)

 |— username1.json # User-specific data files

 └─ username2.json

How to Use

Getting Started

1. Launch the Application

`python health_manager.py`

2. Create an Account

- Select "Register" from the login screen
- Choose a unique username
- Your data file will be created automatically

3. Start Tracking

- Use the main menu to log meals and exercises
- View your records to see progress
- Monitor your net calorie balance

Basic Workflow

Logging a Meal:

1. Select "Log Meal" from main menu
2. Enter food description
3. Input calorie count
4. Choose current or custom date/time
5. Save automatically

Logging an Exercise:

1. Select "Log Exercise" from main menu
2. Enter exercise description
3. Input duration in minutes
4. Enter calories burned
5. Save automatically

Viewing Records:

- View meals, exercises, or all records
- See totals and summaries
- Track your calorie balance

Testing Instructions

Manual Testing Steps

1. User Registration Test

- Register a new user
- Try duplicate username registration
- Test empty username handling

2. Login Test

- Login with valid credentials
- Attempt login with non-existent user
- Verify session persistence

3. Data Logging Test

- Log meals with various calorie values
- Log exercises with different durations
- Test custom date/time entries
- Verify input validation

4. Data Viewing Test

- View meal records
- View exercise records
- Check summary calculations
- Verify sorting functionality

5. Error Handling Test

- Test invalid input handling
- Verify file operation safety
- Check menu navigation errors

Test Data Examples

Sample Meal Entry:

- Food: "Oatmeal with fruits"
- Calories: 350
- Date: 2024-01-15
- Time: 08:30

Sample Exercise Entry:

- Exercise: "Running"

- Duration: 30 minutes
- Calories Burned: 300
- Date: 2024-01-15
- Time: 07:00

Application Screenshots

Login Screen

```
C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe
=====
HEALTH MANAGER APPLICATION
=====

1. Login
2. Register
3. Exit

Choose an option (1-3): 1
```

Main Menu

```
C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe
=====
HEALTH MANAGER APPLICATION
=====
Logged in as: ayushman patri

=== MAIN MENU ===
1. Log Meal
2. Log Exercise
3. View Records
4. Logout
5. Exit Application

Choose an option (1-5): 3
```

Record Summary

```
C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe C:\Users\AYUSH\AppData\Local\Programs\Python\Python312\python.exe
=====
HEALTH MANAGER APPLICATION
=====
Logged in as: ayushman patri

=== ALL RECORDS SUMMARY ===

Total Meals Logged: 2
Total Exercises Logged: 1
Total Calories Consumed: 350
Total Calories Burned: 170
Net Calories: 180
Status: Calorie Surplus

Press Enter to continue...
```

Customization

The application can be easily extended by modifying the health_manager.py file:

- Add new health metrics
- Modify data storage format
- Enhance reporting features
- Add new analysis calculations

Contributing

Contributions are welcome! Please feel free to submit pull requests or open issues for:

- Bug fixes
- New features
- Documentation improvements
- Testing enhancements

Support

If you encounter any issues:

1. Check that Python 3.6+ is installed
2. Verify file permissions in the application directory
3. Ensure no other processes are using the user data files
4. Check the users directory is created and accessible

Future Enhancements

- Data export functionality
- Enhanced visualization
- Meal and exercise templates
- Water intake tracking
- Goal setting features
- Mobile application version